

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination 2023

Course: B. Tech. Branch: Electronics & Telecommunication Engg./Electronics Engg Semester :VI

Subject Code & Name: BTETPE603E/BTEXPE603A Information Theory and Coding

Max Marks: 60

Date: 17/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|-----------|----------|
| A) In a factory, four machines A_1 , A_2 , A_3 and A_4 produce 10%, 20%, 30% and 40% of the items, respectively. The percentage of defective items produced by them is 5%, 4%, 3% and 2%, respectively. An item selected at random is found to be defective. What is the probability that it was produced by the machine A_2 ? | L1 | 6 |
| B) Define CDF. State the properties of CDF. | L1 | 6 |
| C) Write a note on Ergodic process | L1 | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|-----------|----------|
| A) An urn contains 4 white and 3 black balls. Two balls are drawn successively with X denoting the number of black balls.
a. Find the probability function of X
b. Draw the bar chart and histogram. | L2 | 6 |
| B) Write a note on power spectral density and prove its properties. | L1 | 6 |
| C) Explain the Resistor noise with the help of an equivalent diagram and power density spectrum. | L1 | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|--|-----------|----------|
| A) Write an note on joint and conditional entropy. | L3 | 6 |
| B) What is mutual information? Derive its expression from entropies. | L2 | 6 |
| C) Also the Shannon-Fano coding procedure for the following message ensemble: | L2 | 6 |

$[X] = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7]$
 $[Y] = [0.4 \quad 0.2 \quad 0.12 \quad 0.08 \quad 0.08 \quad 0.08 \quad 0.04]$

Take $M=2$.

Q.4 Solve Any Two of the following.

- | | | | |
|----|---|-----------|----------|
| A) | For a systematic linear block code, the three parity check digits are given by
$C_4 = d_1 + d_2 + d_3$
$C_5 = d_1 + d_2$
$C_6 = d_1 + d_3$
i) Construct generator matrix
ii) Construct code generated by this matrix
iii) Determine error correcting capability
iv) Prepare a suitable decoding table
v) Decode the received code words 101100 and 000110 | L3 | 6 |
| B) | Write a note on convolution encoder. Draw its basic structure and explain. | L2 | 6 |
| C) | Write a note on Syndrome decoding. What happens if double error occurs in received code word when the minimum distance criterion is 3? | L3 | 6 |

Q. 5 Solve Any Two of the following.

- | | | | |
|----|---|-----------|----------|
| A) | Discuss any three types of vocoders in brief. | L1 | 6 |
| B) | Write a short note on vector quantization. | L2 | 6 |
| C) | Explain GSM codec. | L2 | 6 |
-